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Locking mechanism used with a flat webbing

Abstract

The present invention is a locking device that can be integrated onto a leash, and which provides a solution to the problem of slip-on collars over tightening or being inconvenient to use. The locking device comprises of a main body, a lock piece, a hook, and one or more arms. The one or more arms are connected within the main body to engage with the hook and the lock piece to lock a width of a collar portion which is formed from the pieces integrated onto the leash. The main body is configured to slide on the leash, and the lock piece is fixed onto the leash and when the lock piece slides into the main body, the main body is prevented from sliding. The hook is connected to a first end of the leash and when inserted into the main body forms the collar portion.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part application which claims priority to U.S. Non-Provisional application Ser. No. 17/685,152 filed on Mar. 2, 2022, which claims priority to U.S. Provisional Patent Application No. 63/155,739 filed on Mar. 3, 2021, which are incorporated by reference in their entirety.

FIELD OF THE DISCLOSURE

(1) The present invention is directed to locking devices, and more particularly to a locking device that can be integrated onto a flat webbing.

BACKGROUND

(2) Dog collars of various types exist. Some collars have a loop that passes over the dog's head and then is tightened. These collars generally take the form of “choker” collars that have no mechanism to stop the collar from tightening or loosening and are held on the dog's neck by keeping tension on the collar. Choker collars can restrict the breathing of a dog wearing the collar with enough tension. Further, if no tension is kept on the choker collar the collar may become loose and move out of position (e.g., over the head of the dog or tangled in the legs of the dog) or even fall off the dog.

(3) Some collars exist that slip over the head of the dog and include mechanisms to stop the collar from becoming too tight. Others include mechanisms that prevent the collar from becoming too loose. However, all of these attempts have been inconvenient for a variety of reasons. One example

of these inconvenient solutions is a collar with a strap that includes holes through which a locking mechanism may pass through to keep the collar from tightening or loosening. This solution is inconvenient because a person putting on the collar must place the locking mechanism in a hole while the collar is around the dog's neck, which may be difficult when a dog is excited to go outside or otherwise is not being compliant, and which is far less convenient than a choker collar which simply slips over the dog's head and is ready to use.

(4) Another example of an inconvenient solution is a collar with a disc or similar device on the strap to prevent the collar from becoming too tight. Although many of these collars can be easily slipped over the head of the dog, the disc does not prevent the collar from becoming too loose or falling off the dog. Also, it can be difficult or impossible to adjust the collar size on the dog because the disc must be fixed to a location on the strap to prevent the collar from tightening.

SUMMARY

(5) One or more embodiments are provided below for a locking device that may be used with a flat webbing material, such as may be used for a leash. The locking device may be integrated onto a flat webbing material to form a slip-on leash for a dog or another animal. In other words, the locking device integrated onto a webbing material may form a leash and collar all in one and that with the locking device the collar can be set to a certain width and then locked to prevent overtightening or loosening. The locking device may also be included on a dog collar that can be slipped around a dog's neck or another animal's neck. The locking device may also be on any other object that can benefit from such a locking device. The locking device may include a main body that is divided into a first section and a second section. One or more arms are connected within the second section. The locking device may also include a locking piece and a hook. The locking piece is configured to fit within the first section of the main body and includes a tab that projects outward from the locking piece which engages with an overhanging portion on one of the arms of the one or more arms to securely lock the locking piece within the main body. The hook is configured to fit within the second section with the one or more arms holding the hook in place within the second section of the main body. The one or more arms may be moved by pressing on to release the locking piece, the hook, or both.

(6) The disclosed device is unique and may provide significant advantages when compared with other devices and solutions known in the arts because it provides a locking device that may be used with a slip-on dog leash or dog collar which prevents the device from tightening or loosening as the leash or collar is being pulled on. The disclosed device also provides a convenient method for putting or removing a slip-on dog leash or dog collar. Additionally, the locking device, when included on a slip-on dog leash or dog collar, allows for convenient adjustments in size of a collar portion of a strap of the device. Such a device may also be integrated onto a flat webbing material that can be used to wrap around an object, in the same manner as discussed above, to provide a possible tying means.

(7) Other advantageous features as well as other aspects and advantages of the invention will be apparent from the following description and the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments of the present disclosure are described in detail below with reference to the following drawings. These and other features, aspects, and advantages of the present disclosure will become better understood with regard to the following description, appended claims, and accompanying drawings. The drawings described herein are for illustrative purposes only of selected embodiments and not all possible implementations and are not intended to limit the scope of the present disclosure.

- (2) FIG. 1 shows an example perspective view of a locking device integrated onto a leash according to one or more non-limiting embodiments.
- (3) FIG. 2 shows a close-up view of collar section of the leash with the locking device according to one or more non-limiting embodiments.
- (4) FIG. 3A shows an exploded view of the locking device according to one or more non-limiting embodiments.
- (5) FIG. 3B shows a close-up view of the locking device in an engaged position according to one or more non-limiting embodiments.
- (6) FIG. 4 shows an example top view of a main body of the locking device according to one or more non-limiting embodiments.
- (7) FIG. 5A shows an example of a lock piece of the locking device according to one or more non-limiting embodiments.
- (8) FIG. 5B shows the lock piece and the leash before the lock piece is fixed to the leash according to one or more non-limiting embodiments.
- (9) FIG. 5C shows the lock piece fixed onto the leash and further shows a projecting element on the lock piece according to one or more non-limiting embodiments.
- (10) FIG. 6A shows a partial cross-sectional view of the locking device in FIG. 3B taken along line B-B illustrating a hook engaging with a first and second arm of the locking device.
- (11) FIG. 6B shows the locking device from FIG. 6A illustrating the hook engaged with the first and second arms of the locking device.
- (12) FIG. 6C shows the locking device from FIG. 6B illustrating the hook disengaged from the first and second arms of the locking device.
- (13) FIG. 7A shows a partial cross-sectional view of the locking device in FIG. 3B taken along line A-A illustrating the lock piece engaging with the first arm.
- (14) FIG. 7B shows the locking device from FIG. 7A illustrating the lock piece disengaged from the first arm.
- (15) FIG. 8 is an example top view of the locking device according to one or more non-limiting embodiments.
- (16) FIG. 9A shows an example view of the collar portion of the leash with the lock piece disengaged from the main body of the locking device.
- (17) FIG. 9B shows the collar portion of the leash integrated with the locking device placed around a dog's neck with the lock piece disengaged from the main body.
- (18) FIG. 9C shows the locking device on the leash in a locked position.
- (19) FIG. 9D shows the hook released from the main body to quickly remove the collar portion of the leash from the dog's neck.
- (20) FIG. 10A shows an example view of an alternative embodiment of a locking device.
- (21) FIG. 10B shows the alternative embodiment of the locking device integrated with a leash and with a hook and a lock piece engaged in a main body of the locking device.
- (22) FIG. 11 is an exploded view of the alternative embodiment of the locking device.
- (23) FIG. 12 shows a top view of a main body of the alternative embodiment of the locking device.
- (24) FIG. 13 shows an internal view of the alternate embodiment of the locking device illustrating the hook engaged with the first and second arms.
- (25) FIG. 14 shows an internal view of the alternate embodiment of the locking device illustrating the lock piece engaged with the first and second arms.
- (26) FIG. 15 shows a perspective view of yet another alternative embodiment of a locking device.
- (27) FIG. 16 shown an exploded view of the alternative embodiment of the locking device from FIG. 15.
- (28) FIG. 17 shows a top perspective view of a main body of the alternative embodiment of the locking device of FIG. 15.
- (29) FIG. 18 shows a top view of the main body and a first arm and a second arm of the locking

device of FIG. 15.

(30) FIG. 19 shows a front perspective of the a lock piece of the locking device of FIG. 15.

(31) FIG. 20 is a cross-sectional view of the locking device taking along line A-A from FIG. 15.

(32) FIG. 21 is a cross-sectional view of the locking device taking along line B-B from FIG. 15.

(33) FIG. 22 illustrates the first arm and the second arm with their respective projections.

DETAILED DESCRIPTION

(34) In the Summary above, in this Detailed Description, the claims below, and in the accompanying drawings, reference is made to particular features of the invention. It is to be understood that the disclosure of the invention in this specification includes all possible combinations of such particular features. For example, where a particular feature is disclosed in the context of a particular aspect or embodiment of the invention, or a particular claim, that feature can also be used—to the extent possible—in combination with and/or in the context of other particular aspects and embodiments of the invention, and in the invention generally.

(35) The term “comprises” and grammatical equivalents thereof are used herein to mean that other components, ingredients, steps, etc. are optionally present. For example, an article “comprising” (or “which comprises”) components A, B, and C can consist of (i.e., contain only) components A, B, and C, or can contain not only components A, B, and C but also contain one or more other components.

(36) Where reference is made herein to a method comprising two or more defined steps, the defined steps can be carried out in any order or simultaneously (except where the context excludes that possibility), and the method can include one or more other steps which are carried out before any of the defined steps, between two of the defined steps, or after all the defined steps (except where the context excludes that possibility).

(37) The term “at least” followed by a number is used herein to denote the start of a range including that number (which may be a range having an upper limit or no upper limit, depending on the variable being defined). For example, “at least 1” means 1 or more than 1. The term “at most” followed by a number is used herein to denote the end of a range, including that number (which may be a range having 1 or 0 as its lower limit, or a range having no lower limit, depending upon the variable being defined). For example, “at most 4” means 4 or less than 4, and “at most 40%” means 40% or less than 40%. When, in this specification, a range is given as “(a first number) to (a second number)” or “(a first number)-(a second number),” this means a range whose limits include both numbers. For example, “25 to 100” means a range whose lower limit is 25 and upper limit is 100 and includes both 25 and 100. Certain terminology and derivations thereof may be used in the following description for convenience in reference only and will not be limiting. For example, words such as “upward,” “downward,” “left,” and “right” would refer to directions in the drawings to which reference is made unless otherwise stated. Similarly, words such as “inward” and “outward” would refer to directions toward and away from, respectively, the geometric center of a device or area and designated parts thereof. References in the singular tense include the plural, and vice versa, unless otherwise noted.

(38) The term “coupled to” as used herein may mean a direct or indirect connection via one or more components.

(39) Referring now to the drawings and the following written description of the present invention, it will be readily understood by those persons skilled in the art that the present invention is susceptible to broad utility and application. Many embodiments and adaptations of the present invention other than those herein described, as well as many variations, modifications, and equivalent arrangements will be apparent from or reasonably suggested by the present invention and the detailed description thereof, without departing from the substance or scope of the present invention. This disclosure is only illustrative and exemplary of the present invention and is made merely for purposes of providing a full and enabling disclosure of the invention.

(40) FIG. 1 shows a perspective view of an example of a non-limiting embodiment of a locking

device **100** integrated onto a leash **400**, which is a type of flat webbing, and that is adapted to be slipped onto an animal's neck, such as a dog. The locking device **100** can be integrated onto any flat webbing material and can be used together with the locking device **100** to wrap around an object. For the remainder of this disclosure, the leash **400** will be used as an example of a webbing material that is integrated with the locking device **100**. Also, for the remainder of this disclosure, the leash **400** and locking device **100** will be used with a dog, as an example. The locking device **100** is integrated onto a leash **400** to form a collar section **410** and a strap section **420**. The collar section **410** of the leash **400** forms a loop that is placed around the dog's neck. The collar section **410** further includes an adjustment piece **412**, such as and not limited to a buckle or a slide glider, that allows the collar section **410** to widen or shorten. The strap section **420** of the leash may include a handle **422**. The leash **400** may be similar to other leashes that are known and available in the arts or will be in the future. The leash **400** integrated with the locking device **100** can come in many sizes and lengths to accommodate small, medium, and large sized dogs. The leash **400**, for example, may have a length of about 5-8 feet. In some example embodiments, the leash **400** may be parts of a single continuous webbing strap, cord, chain, rope, or other similar material. In other embodiments, portions of the leash **400** may be made of different materials.

(41) The handle **422** and the collar portion **410** are formed on opposite ends of the leash **400** with a length of the strap **420** separating the two. The leash **400** may have a first end **401** and a second end **402**. The collar portion **410** is formed near the first end **401**. The handle **422** is formed near the second end **402**, which may be formed by doubling over the strap **400** and attaching (by sewing, glue, rivet, etc.) the second end **402** of the leash **400** to another portion of the strap **422** to form a loop sized for a human hand. The adjustment piece **412** is attached to the leash **400** within the collar section **410**. As mentioned above, the adjustment piece **412** allows the collar section **410** to widen or shorten the loop to fit around a dog's neck. It is to be understood that this is an example of how the leash **400** may be formed, however, there are variations to other forms and shapes that the leash **400** may take.

(42) Refer to FIGS. 2, 3A, and 3B regarding the locking device **100** integrated on to the leash **400**. FIG. 2 is close-up view of the collar section **410** of the leash **400** with the locking device in an unengaged position. FIG. 3A is an exploded view of the locking device **100**. FIG. 3B is a close-up view of the locking device **100** in a fully engaged position. As shown in FIG. 2, the locking device **100** is integrated onto the leash **400** and the placement of the different elements of the locking device **100** assist in forming the collar section **410** of the leash **400**. The locking device **100** comprises of a main body **110**, a hook, **120**, a lock piece **130**, and one or more arms **140**, **142**. The hook **120** is connected to the first end **401** of the leash **400**. The lock piece **130** is fixedly connected to the leash **400** and is positioned between the adjustment piece **412** and the main body **110** which is positioned on the leash **400** between the lock piece **130** and the handle **422**. The adjustment piece **412** of the leash is between the hook **120** and the lock piece **130**. In other words, when looking at the leash **400** from the first end **401** to the second end **402**, the hook **120** is connected to the first end **401**, next is the adjustment piece **412** on the leash **400**, next is the locking piece **130** which is fixed onto the leash, then the main body **110** of the locking device **100**, then the strap section **420**, and lastly the handle **422** configured from looping the second end **402**.

(43) FIG. 2 illustrates that the lock piece **130** is fixedly connected to the leash **400**, and the leash **400** may then pass through a slit **112a** (shown in FIG. 4) in the main body **110** on a side that moves toward the second end **402** which has the handle **422**. From the lock piece **130**, on a side moving toward the first end **401**, the leash **400** may pass through the adjustment piece **412** (with the fixed lock piece **130** positioned between the main body **110** and the adjustment piece **412**). From the adjustment piece **412**, the leash **400** may then pass through an opening **122** in the hook **120** of the locking device **100** and loop back to the adjustment piece **412**. Because a section of the collar portion **410** between the hook **120** and the adjustment piece **412** is a looped section **414**, the adjustment piece **412** can be used to adjust the length of the collar portion **410**, which will be

discussed below.

(44) The strap portion **420** of the leash **400** may slide through the slit **112a** (See, FIG. **4**) in the main body **110** of the locking device **100**, except when the lock piece **130** is engaged with the main body **110**, which will be discussed in more detail below. In other words, the main body **110** is not fixed to the leash **400** such that the main body **110** may slide along the strap portion **420** between the handle and the lock piece **130**. When the lock piece **130** is engaged with the main body **110**, the main body **110** is prevented from sliding on the leash **400**. When the lock piece **130** is not engaged with the main body **110**, the collar portion **410** may be widened allowing for the collar portion **410** to be placed over the head of a dog. Then, the lock piece **130** may be slid toward the main body **110** by pulling on the strap section **420** to engage the lock piece **130** with the main body **110** to lock and prevent further movement of the main body **110** on the leash **400** and lock a size of the collar portion **410**. The one or more arms **140**, **146** are connected to the main body **110** and may be used to engage and disengage the lock piece **130** from the main body **110**. Additionally, the one or more arms **140**, **142** may also be used to engage and disengage the hook **120** from the main body **110**. The hook **120** when engaged is attached to the main body **110**, and when disengaged can be removed from the main body **110**, allowing the collar portion **410** of the leash **400** to fully open and undo the loop of the collar portion **410**. Removing the hook **120** from the main body **110** allows the collar portion **410** to come off the dog's neck quickly.

(45) The size of the loop of the collar portion **410** may be adjusted using the adjustment piece **412**. The looped section **414**, and a straight section **416** (a section between the lock piece **130** and the adjustment piece **412**) together determine the length of the loop of the collar portion **410** (when the lock piece **130** is engaged with the main body **110**). By moving the adjustment mechanism **412**, the looped section **414** can be lengthened or shortened, and this changes the size of the loop of the collar portion **410**. The size of the loop of the collar portion may be increased (and conversely the length of the looped section **414** can be shortened) when the adjustment mechanism **412** is slid away from the lock piece **130** and toward the hook **120** of the locking device **100**. The size of the loop of the collar portion **410** may be decreased (and conversely the length of the looped section **414** may be increased) when the adjustment piece **412** is moved toward the lock piece **130** and away from the hook **120** of the locking device **100**.

(46) Referring specifically to the locking device **100**, FIGS. **3A** and **3B** illustrate the elements that comprise the locking device **100** and how the key components are connected to each other to provide a useful functionality. Also referring to FIG. **4**, a top view of a non-limiting example of the main body **110** is shown. The main body **110** provides a structural support for the remainder of the elements comprising the locking device **100**. The main body **110** is configured with four sides connected to a closed bottom end and having an open top end. The terms “bottom” and “top” are strictly being used for reference purposes. The main body **110** is further configured with a first section **112** and a second section **114** separated by a center wall **116**. The main body may be fashioned from a durable material including and not limited to plastic, metal, a composite, or other durable materials suited for withstanding the forces exerted by a dog pulling on a leash attached to the locking device **100**.

(47) The first section **112** and the second section **114** are configured to accept the lock piece **130** and the hook **120**, respectively. The first section **112** has the slit **112a** configured on the bottom end of the locking device **100**. As mentioned above, the lock piece **130** is fixedly connected to the leash **400** and the slit **112a** is adapted to allow the leash **400** to pass through when the locking device is attached to the leash **400**. In operation, the leash **400** can slide back and forth through the slit **112a** as the main body **110** is not fixed to the leash **400**. In other words, the main body **110** is connected to the leash **400** by the leash **400** being inserted through the slit **112a**. The leash **400** will slide through back and forth till the lock piece **130** slides into the first opening **112** from the open top end and is locked into place. The slit **112a** has a width that allows the leash **400** to slide back and forth but not wide enough for the lock piece **130** to go through. In other words, the lock piece **130**

abuts against the bottom end of the main body **110** which includes the slit **112a** and allows the lock piece **130** to nest in first section **112** of the main body **110**. The nesting of the lock piece **130** in the first section **112** of the main body **110** keeps the collar portion from tightening any further, so the width of the collar portion is set when the lock piece **130** is nesting in the main body **110**.

(48) The second section **114** is configured to include the one or more arms **140**, **142**. The one or more arms **140**, **146** are attached to the main body **110** within the second section **114**. As clearly seen in FIG. 3A, the second section **114** is configured with a depression **115** on the side defining part of the second section and which is parallel to the center wall **116**. The depression **115** is included to allow the hook **120** to be inserted into the second section **114** and sit flush against the top side and further prevent any side-to-side movement of the hook **120**, as shown in FIG. 3B. The second section further includes openings **117** that traverse the side of the main body **110** which defines part of the second section **114** and includes openings **117** in the center wall **116**. The openings **117** on the side of the main body **110** and the center wall **116** are aligned together on the same plane so that the one or more arms **140**, **146** may be to be connected to the main body **110** within the second section **114**, which will be discussed below.

(49) The first section **112** and the second section **114** of the main body **110** may be rectangular in shape. A rectangular shape is well suited for the leash **400** integrated with the lock device **130** that includes a flat shape for the leash **400**. The first section **112** and second section **114** may have other shapes if the leash that passes through the first section **112** is a rope, chain, cable, etc. It is to be understood that the first section **112** and the second section **114** may have different shapes wherein the second section **114** may have a rectangular shape and the first section may have an alternate shape designed to accommodate the shape of the leash **400**.

(50) The lock piece **130** is depicted in FIGS. 3A-B and 5A-C. In FIG. 3A, the lock device **130** is shown to be integrated onto the leash **400** which is inserted into the slit **112a** in the first section **112** of the main body **110**. In FIG. 3B, the lock device is inserted into the first section and engaged with the one or more arms **140**, **146**, which will be discussed below. In FIGS. 5A-5C, the lock device **130** is shown to be comprised of two pieces, a first lock piece **130a** and a second lock piece **130b** which are fixed onto the leash **400** with the leash **400** between the first and second lock pieces **130a**, **130b**. As best seen in FIG. 5A, an inside surface of the first and second lock pieces **130a**, **130b** are shown. FIG. 5B shows that the inside surfaces of the first and second lock pieces **130a**, **130b**, face each other and attach to the leash **400**. In some embodiments, the inside surfaces of the first and second lock pieces **130a**, **130b** are configured with a plurality of spikes **132**, **134**, respectively. One example of the spikes **132**, **134**, may be an arrowhead type appearance with a pointed tip. The spikes are useful to pierce into the leash **400** to partially secure the lock piece **130** to the leash **400**. It is to be understood that the first and second lock pieces **130a**, **130b**, may not include the spikes **132**, **134**.

(51) The first and second lock pieces **130a**, **130b** are attached to the leash **400** by using a securing element. Referring to FIGS. 5A-C, in one or more embodiments, the first and second lock pieces **130a**, **130b**, may have a plurality of holes **136** that traverse through each lock piece **130**. The holes **136** on the first lock piece **130a** and the second lock piece **130b** align with each other when each lock piece **130** is placed onto the leash **400**. The plurality of holes **136** allow the first and second lock pieces **130a** and **130b** to be fixed to the leash **400** and to each other. The plurality of holes **136** are particularly useful for using a fastener **137** to be placed through to attach the lock pieces **130** together to the leash **400**. An example of a fastener **137** may include rivets. It is also to be understood that the first and second pieces **130a**, **130b**, may be fixed to each other and the leash **400** by using other securing elements such as and not limited to thread, glue, staples, or another securing element.

(52) FIG. 5C shows an example side view of the lock device **130** on the leash **400**. In this Figure, only the first lock piece **130a** of the strap **100** is shown to allow for greater detail to be shown on the described portion. The first lock piece **130a** has a projecting element **138** on an outward facing

surface (surface opposite the side fixed to the leash **400**) which is shaped to engage with the one or more arms **140**, **142** to lock the locking device **130** into the main body **110**. The first lock piece **130a** with the projecting element **138** when slid into the first section **112** of the main body **110** enters facing the center wall **116**. The center wall **116** has a groove **118** that is open from the top end of the main body **110** to the bottom end of the main body **110**. The projecting element **138**, as the name suggests, projects outward from the first lock piece **130a**, and as the lock device slides into the first element **112**, the projecting element is guided into the groove **118** of the center wall **116** (See, FIG. 7A).

(53) As discussed earlier, the one or more arms **140**, **142**, are connected to the main body **110** within the second section **114**. The one or more arms **140**, **142**, may be configured to cause the lock piece **130** to engage and disengage within the main body **110** of the locking device **100**.

Additionally, the one or more arms **140**, **142**, may also be configured to cause the hook **120** to engage and disengage within the main body **110**. As seen in the Figures, the locking device **100** has two arms, a first arm **140** and a second arm **142**. Each of the first and second arms **140**, **142**, are attached to the main body **110** within the second section **114** by a pin **145** which is inserted into the openings **117** in the second section and the openings **141**, **143**, on the first and second arms **140**, **142**, respectively. The pin **145** fixes each of the first and second arms **140**, **142** to the first section **114** of the main body. A portion of the first and second arms **140**, **142** extends outside of the main body **110**, which allows a surface for a user to press on to move the first and second arms **140**, **142** at the pin **145** which functions as an axel. Stated another way, the first and second arms **140**, **142** rotate at the pin **145** within the second section **114** of the main body **110**.

(54) The first and second arms **140**, **142** extend toward the bottom end of the main body **110** and a tip of each arm **140**, **142** at the bottom end are relatively close to each other or almost touching. Referring to FIG. 7A the first and second arms **140**, **142**, may each include a spring engaging portion **140a**, **142a**, respectively. Specifically, the spring engaging portions **140a**, **142a** are relatively configured near the tip of each arm **140**, **142**, and facing toward the side of the main body **110**. Each of the spring engaging portions **140a**, **142a**, are attached to a spring device **144**, **146**, respectively, which push against an inside wall of the second section **114** in the main body **110**. The spring engaging portions **140a**, **142a** attached to their respective spring device **144**, **146**, provide a restoring force on the first and second arms **140**, **142**, respectively. A user applies pressure (by pressing) on the first and second arms **140**, **142**, rotating the first and second arms at the pin **145**. The rotation with the applied pressure moves the first and second **140**, **142**, upward at the outside portion and the tips away from a center of the bottom end of the second section **114**. Releasing the pressure off the first and second arms **140**, **142**, returns the first and second arms **140**, **142**, back to their resting position which is with the tips near the center of the bottom end of the second section **114** of the main body **110**. The spring devices **144**, **146** may be steel springs or other similar devices that provide a restoring force. The first and second arms **140**, **142** may be fashioned from the same material as the main body **110** including and not limited to metal, plastic, or other durable materials.

(55) FIGS. 6A to 6C is a cross-sectional view taken along line B-B in FIG. 3B to depict the hook **120** engaging and disengaging with the first and second arms **140**, **142**. Disengaging the hook **120** detaches the hook from the main body **110** and opens up the collar portion **410**. In other words, the detaching the hook **120** from the main body undoes the collar portion **410**. The hook **120** has a broad tip **124** which enters the second section **114** and engages with the first and second arms **140**, **142**. FIGS. 6A and 6B show that as the hook **120** is inserted into the second section **114**, the broad tip **124** pushes against the first and second arms **140**, **142**, causing them to move backwards to make space for the broad tip **124** on the hook **120** to move past the first and second arms **140**, **142** to fit at the bottom end of the second section **114**. Once the broad tip **124** on the hook **120** moves past the first and second arms **140**, **142**, the spring devices **144**, **146** move the first and second arms **140** back to their original position engaging them over the broad tip **124** and preventing the hook

120 from moving. Thus, the first and second arms **140**, **142** lock the hook **120** into the second section **114**. FIG. **6C** shows that the hook **120** is disengaged and unlocked when the first and second arms **140**, **142** are moved at the pins **145** causing the first and second arms **140**, **142** to move away from the broad tip **124** freeing the hook **120** to slide out of the second section **114**. (56) FIGS. **3A** and **8** also illustrate that the first arm **140** has a wide tip **148** which extends toward the center wall **116**, and more particularly toward the groove **118** in the center wall **116**. The wide tip **148** locks the lock piece **130** in the first section **112** of the main body **110**. Specifically, the wide tip **148** of the first arm **140** engages with the projecting element **138** on the lock piece **130** to hold the lock piece **130** within the first section **112** of the main body **110**. FIGS. **7A** and **7B** show example cross-sectional views taken from FIG. **3B** along line A-A. FIG. **7A** and FIG. **8** show the first and second arms **140**, **146** connected within the second section **114** with a view of the lock device **130** within the first section (not shown in this figure). The projecting element **138** is shown in the groove **118** of the center wall **116**. The first arm **140** is shown to be engaged with the lock piece **130** by preventing the movement of the lock piece out of the first section of the main body **110**. Specifically, the wide tip **148** is above the projecting element **138** in the groove of the main body **110** such that the lock piece **130** is prevented from moving upward by the projecting element **138** prevented from moving upward in the groove **118** by the wide tip **148**. FIG. **7B** shows the first arm **140** disengaged from the lock piece **130**. Specifically, the first arm **140** may be moved at the pin **145** causing the wide tip **148** to be moved away from a position above the projecting element **138** and thus allowing the lock device **130** to move upward with the projecting element **138** moving upward through the groove **118** unhindered.

(57) FIGS. **9A** to **9D** depict an embodiment of the locking device **100** integrated onto a leash **400** being used with a dog **500**. In FIG. **9A**, a user would ensure that the lock piece is disengaged and not inserted into the main body **110** of the locking device. The user may widen the collar portion **410** of the leash **400** to be able to slip it over the dog's head. FIG. **9B** shows that the collar portion **410** has been slipped over the dog's head and is loosely placed around the dog's neck. In FIG. **9C**, the strap section **420** of the leash **410** may be pulled on to slide the leash through the main body **110** moving the lock piece **130** into the main body **110** to engage with the first arm locking the lock piece into place. Once locked, the width of the collar portion **410** is set and as the leash is pulled on while around the dog's neck, the collar portion will not tighten or loosen. FIG. **9D** shows where the hook **120** can detach from the main body **110** for a quick release of the locking device from the dog without needing to slip the collar portion **410** over the dog's head. This may be useful for speedily releasing the dog from the leash **400**.

(58) The locking device **100** integrated onto a leash **400** has several advantages over existing dog collars. The locking device **100** may be slipped over the head of the dog and secured by sliding the main body **110** along the leash until the lock piece **130** fixed onto the leash **400** enters the main body **110**. Thus, the locking device **100** integrated onto a leash may be more easily and more conveniently placed over on a dog. The lock piece **130** engages with the first arm to prevent the collar portion **410** placed around the dog's neck from loosening by preventing the lock piece from sliding back out of the open top end. Additionally, the lock piece **130** is able to nest within the main body by pushing against the closed bottom end and thus the collar portion is prevented from tightening an further on the dog's neck. The size of the collar portion **410** can be adjusted using the adjustment mechanism **412** so the leash **400** integrated with the locking device **100** can be adjusted for dogs of different sizes. The quick release mechanism of the hook **120** allows the leash **400** integrated with the locking device **100** to be quickly and easily released from around the dog's neck.

(59) FIGS. **10A** to **14** depict an alternative embodiment of a locking device **200** that may be integrated onto a leash **400** that can be secured around the neck of a dog similar to the disclosure above in reference to the first embodiment of the locking device **100** of FIG. **1**. FIG. **10A** illustrates the locking device **200** only in order to provide a clear view of all the elements that comprise the

locking device **200**. FIG. **10B** illustrates the locking device **200** integrated with leash **400** to show that the like elements of the locking device **200** are integrated onto the leash **400** as discussed above in reference to the locking device **100** integrated onto the leash **400**. FIG. **11** illustrates an exploded view of the locking device **200** only to provide a clearer view of the elements comprising the locking device **200**.

(60) The locking device **200** comprises of similar elements to the locking device **100** including a main body **210**, a hook **220**, a lock piece **230**, and first and second arms **240**, **242**. All elements are the same as discussed in the above embodiment of the locking device **100**. There are a few differences which will be discussed herein. As best seen in FIGS. **11** and **12**, the main body **210** has a center wall **216** with two parallel grooves **218** separated by a center post **216a** (whereas the locking device **100** discussed above has one groove **118**). The first and second arms **240**, **242** both engage with the lock piece **230**, and both the first and second arms **240**, **242** are configured with a wide tip **248**, **249**, respectively as seen in FIGS. **11** and **14**. Thus, as seen in FIG. **14**, the lock piece **230** is configured with two projecting elements **238** wherein each of the projecting elements is guided into one groove so that each of the two grooves has one of the projecting elements **238** side by side as the lock piece is inserted into the first section. The two projecting elements **238** engage with the wide tips **248**, **249**, on the first and second arms **240**, **242**, respectively. The first and second arms **240**, **242**, also engage with the hook **220**. The process to engage and disengage the hook **220** is the same as discussed above for the locking device **100**. However, both the first and second arms **240**, **242**, need to be pressed on to move the arms **240**, **242** away from the projecting elements **238** on the lock piece to allow the lock to be moved out of a first section **212** of the main body **210**. The first and second arms **240**, **242** need only be pressed halfway to move them away from the projecting elements **238** to disengage and move the lock piece **230**. Fully depressing the first and second arms **240**, **242**, moves them away from a broad tip **224** of the hook **220** and thus allowing both the hook **220** and the lock piece **230** to disengage and move out of the main body **210**.

(61) FIGS. **15** to **22** depict yet another alternative embodiment of a locking device **300** that may be integrated onto a leash **400**, as shown in the previous two embodiments. This non-limiting embodiment of the locking device **300** can be used with the leash **400**, similar to the disclosure above in reference to the first two embodiments of the locking device **100**, **200** of FIG. **1** and FIG. **10**, respectively. FIG. **15** is a side perspective view of the locking device **300** to illustrate how elements comprising the locking device **300** are put together. The locking device **300** is not shown integrated to a leash **400**, however, the locking device **300** would be integrated with a leash similarly to how the locking devices **100**, **200** are integrated with their respective leash **400**. The integration of the leash **400** is discussed above in reference to the locking devices **100**, **200**. FIG. **16** illustrates an exploded view of the locking device **300** to provide a clearer view of the elements comprising the locking device **300**.

(62) The locking device **300** comprises of similar elements to the locking device **200** including a main body **310**, a hook **320**, a lock piece **330**, and first and second arms **340**, **342**. All elements are the same as discussed in the above embodiment of the locking devices **100**, **200** discussed in FIGS. **1** to **14**. Like numbers denote like elements which include the same feature and functionality. There are a few differences which will be discussed herein. The main body **310** has a center wall **316** with a central groove **318**, which includes a knob **316a** protruding from a relative center of the central groove (whereas the locking device **200** has the central groove **218** with a post **216a** extending from the center of the central groove **216**; and whereas the locking device **100** discussed above has one groove **118**).

(63) As seen in FIG. **19**, the lock piece **330** comprises of two pieces, a first lock piece **330a** and a second lock piece **330b**, wherein the first lock piece includes a projecting element **338** with a channel **338a** on a bottom side of the projecting element. The projecting element **338** slides into the central groove **318** of the main body **310** and the channel **338a** is guided onto the knob **316a** which

ensures that the lock piece **330** is positioned properly into a first section **312** of the main body **330**. (64) The first and second arms **340**, **342** are configured to engage with the lock piece **330** and the hook **320**. The first and second arms **340**, **342** are also configured such that both the first and second arms **340**, **342** have reduced individual movement essentially keeping each from being pushed individually and only allowing the first and second arms **340**, **342** to move together. As best seen in FIGS. **16** and **18**, both the first and second arms **340**, **342** are configured with a wide tip **348**, **349**, respectively, which are like the wide tips **248**, **249** on the first and second arms **240**, **242** of the locking device **200** The first and second arms **340**, **342** engage with the lock piece **330** by having the wide tips **348**, **349** resting against a top of the projecting element **338**. FIG. **21** is a cross sectional view of the locking device **300** taking along the line B-B to show the wide tips **348**, **349** resting atop the projecting element **338**.

(65) The first and second arms **340**, **342** also engage with the hook **320**. The process to engage and disengage the hook **320** is the same as discussed above for the locking device **100**. Once the broad tip **324** on the hook **320** moves past the first and second arms **340**, **342** into the main body **310**, spring devices **344**, **346** on the first and second arms function similarly to the spring devices **144**, **146** on the locking device **100**. The spring devices **344**, **346** move the first and second arms **340**, **342** back to their original position engaging them over the broad tip **324** and preventing the hook **320** from moving. Similar to the second embodiment of the locking device **200**, both the first and second arms **340**, **342**, need to be pressed on to move the arms **340**, **342** away from the projecting element **338** on the first lock piece **330a** to allow the lock piece **330** to be moved out of the first section **312** of the main body **310**. The first and second arms **340**, **342** need only be pressed halfway to move them away from the projecting element **338** to disengage and move the lock piece **330**. Fully depressing the first and second arms **340**, **342**, moves them away from a broad tip **324** of the hook, allowing both the hook **320** and the lock piece **330** to disengage and move out of the main body **310**. FIG. **20** illustrates a cross sectional view of the locking device taken along line A-A to illustrate the engagement of the first and second arms **340**, **342** with the broad tip **324** of the hook **320**.

(66) However, as mentioned above, the first and second arms **340**, **342** are configured to restrict individual movement and only allow them to move together. The first and second arms **340**, **342** may allow a few degrees of individual movement such that the arms **340**, **342** are pressed on simultaneously, they may move out of each other's way to fully disengage from the lock piece **330**. As best seen in FIGS. **16**, **18** and **22**, the first arm **340** has a projection, referred to as a first arm projection **350**, which extends relatively perpendicular from a top surface of the first arm and is substantially a length of the top surface. The first arm projection **350** also includes an inset portion **351**, and a hook portion **352**. The second arm **342** also includes a projection, a second arm projection **355**, which extends perpendicularly from a top surface of the second arm **342**, wherein the second arm projection **355** has an end **356** that is configured to catch into the inset portion **351** of the first arm projection **350**. The second arm projection **355** also has an extension portion **357** at a bottom end of the second arm **342** which is configured to catch onto the hook portion **352** of the first arm projection **350**. In this way, the first and second arm projections **350**, **355** restrict the individual movement of the arms **340**, **342** to only a few degrees, such as but not limited to, movement between 5 to 12 degrees.

(67) Referring to FIG. **18**, it can be seen that the first arm and second arm projections **350**, **355** extend from a portion of the first arm **340** and a portion of the second arm **342** such that a remainder of the top surface of the first arm and a remainder of the top surface of the second arm do not have any projection and are smooth. When the first and second arms **340**, **342** are placed within a second section **314** of the main body **310**, the first and second arm projections **350**, **355** abut against the center wall **316**. Thus, these projections **350**, **355** are not in the way of the hook piece **320** entering the second section **314** which would slide over the smooth portion of the first and second arms **340**, **342** when inserted into the second section **314** of the main body **310**. The

hook piece **320** functions similar to the hook piece **120** described in the embodiment of the locking device **100**. Thus, like the locking device **100** with the hook **120**, the hook **320** of the current embodiment of the locking device **300** is inserted into the second section **314**, a broad tip **324** pushes against the smooth portion of the first and second arms **340**, **342**, causing them to move backwards to make space for the broad tip **324** to move past the first and second arms **340**, **342** to fit at the bottom end of the second section **314**. However, the hook **320** has a hinge section **321**, which is a section that rests in a depression **315** in the second section **314**. The depression has a similar function as that described in the above embodiment of the locking device, which is included to guide the hook **320** to be inserted properly into the second section **314** and prevent side to side movement of the hook **320**. The hinge section **321** reduces non-linear tension on the hook **320** and allows the hook **320** to move linearly as a connected leash is pulled on.

(68) Accordingly, the present description provides for various embodiments for a locking device that can be integrated onto a flat webbing material that can be used to secure the flat webbing material around an object. An example use may include integrating the locking device onto a leash material that can be secured around the neck of a dog or other animal. Many uses and advantages are offered by the locking device as described above in one or more non-limiting embodiments in the present description.

(69) The corresponding structures, materials, acts, and equivalents of any means or step plus function elements in the claims below are intended to include any structure, material, or act for performing the function in combination with other claimed elements as specifically claimed. The description of the present invention has been presented for purposes of illustration and description but is not intended to be exhaustive or limited to the invention in the form disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the invention.

(70) The embodiments were chosen and described in order to best explain the principles of the invention and the practical application, and to enable others of ordinary skill in the art to understand the invention for various embodiments with various modifications as are suited to the particular use contemplated. The present invention, according to one or more embodiments described in the present description, may be practiced with modification and alteration within the spirit and scope of the appended claims. Thus, the description is to be regarded as illustrative instead of restrictive of the present invention.

Claims

1. A device comprising: a main body having four sides, a center wall, a closed bottom, and an open top, wherein the center wall divides the main body into a first section and a second section; a first arm and a second arm connected to the main body; a hook configured to slide into the main body and engage with the first arm and the second arm within the main body, wherein the hook has a hinged section; a lock piece configured to slide into the main body to nest against the closed bottom and engage with the first arm and the second arm; and wherein the hook and the lock piece are fixedly connected to a flat webbing material and the main body has a slit in the closed bottom of the first section, wherein the flat webbing material is inserted through the slit, so the main body is slidably connected to the flat webbing material, wherein the main body is prevented from sliding when the lock piece slides into the main body.

2. The device of claim 1, wherein the hook removably connects to the lock piece to form a collar portion and a strap with the flat webbing material, wherein an adjustment piece between the hook and the lock piece is configured to adjust a length of the collar portion.

3. The device of claim 1, wherein the hook has an opening configured to allow the flat webbing material to fit through and loop around the opening to connect the flat webbing material to the hook, and wherein the hinged section fits within a recessed portion of the main body.

4. The device of claim 1, wherein the lock piece has a first piece and a second piece which are fixedly connected to the flat webbing material, wherein the flat webbing material is positioned between and fixedly connected to the first piece and the second piece by one or more securing elements.

5. The device of claim 4, wherein a surface of the first piece and a surface of the second piece which are placed against the leash material are each configured with a plurality of spikes.

6. The device of claim 2, wherein the collar section is formed with the lock piece and the hook engaged with the first and second arms in the main body, and wherein the lock piece is nesting within the first section and the hook is inserted in the second section.

7. The device of claim 1, wherein the first arm and the second arm are connected to the main body by a pin which is placed within openings that traverse through the first and second arms, a side of the main body, and the center wall defining the second section, wherein the first and second arms are configured to move at the pin, and wherein the first and second arms each have a spring engaging portion and a connected spring for moving the first and second arms back to their original positions when they are moved at the pins.

8. The device of claim 1, wherein the hook is configured with a broad tip which engages with the first and second arms when the hook is inserted into the second section of the main body; wherein the first arm and the second arm are engaged with the hook by resting above the broad tip to prevent the hook from sliding out of the second section; and wherein the hook is disengaged when the first and second arms are moved away from the broad tip of the hook freeing the hook to slide out of the second section.

9. The device of claim 6, wherein a lower end of the lock piece, facing toward a center wall, includes a projecting element having a channel, and wherein the center wall has a central groove with a knob included on the closed bottom within the central groove, wherein the projecting element of the lock piece slides into the central groove with the knob accommodating the channel of the projecting element when the lock piece slides into the first section.

10. The device of claim 9, wherein the first arm and the second arm each have a wide tip wherein each of the wide tips extends toward the central groove in the center wall to engage with the projecting element, wherein the wide tips extending above the projecting element prevent the lock piece from moving out of the first section; and wherein the lock piece is disengaged when the wide tips of the first arm and the second arm are moved away from the projecting element freeing the lock piece to slide out of the first section.

11. A device comprising: a main body having a first section and a second section divided by a center wall, wherein a top end is open, and a bottom end is closed, and the first section has a slit in the bottom end; a first arm and a second arm connected to the main body, wherein the first arm includes a first arm projection extending from a portion of a top surface of the first arm, and the second arm includes a second arm projection extending from a portion of a top surface of the second arm a hook configured to slide into the main body to engage with the first arm and the second arm within the main body; a lock piece configured to slide into the main body to nest against the closed bottom and engage with the first arm and the second arm; and wherein the hook and the lock piece are fixedly connected to a flat webbing material, wherein the flat webbing material is inserted through the slit in the first section, so the main body is slidably connected to the flat webbing material, wherein the main body is prevented from sliding when the lock piece slides into the main body.

12. The device of claim 11, wherein: the first arm projection extends from the portion of the top surface of the first arm such that the first arm projection abuts against the center wall and a remainder of the top surface of the first arm is smooth; the second arm projection extends from the portion of the top surface of the second arm such that the second arm projection abuts against the center wall and a remainder of the top surface of the second arm is smooth; and the first arm projection and the second arm projection engage with each other to restrict individual movement of

the first arm and the second arm.

13. The device of claim 11, wherein the hook removably connects to the lock piece to form a collar portion and a strap with the flat webbing material, wherein an adjustment piece between the hook and the lock piece is configured to adjust a length of the collar portion.

14. The device of claim 13, wherein the collar portion is formed with the lock piece and the hook engaged with the first and second arms in the main body, and wherein the lock piece is nesting within the first section and the hook is inserted in the second section.

15. The device of claim 11, wherein the hook has an opening configured to allow the flat webbing material to fit through and loop around the opening to connect the flat webbing material to the hook, and wherein the hook further includes a hinged section that fits within a recessed portion of the main body.

16. The device of claim 11, wherein the lock piece has a first piece and a second piece which are fixedly connected to the leash material, wherein the leash material is positioned between and fixedly connected to the first piece and the second piece by one or more securing elements, and wherein a surface of the first piece and a surface of the second piece which are placed against the leash material are each configured with a plurality of spikes.

17. The device of claim 11, wherein the first arm and the second arm are connected to the main body by a pin which is placed within openings that traverse through the first and second arms, a side of the main body, and the center wall defining the second section, wherein the first and second arms are configured to move at the pin, and wherein the first and second arms each have a spring engaging portion and a connected spring for moving the first and second arms back to their original positions when they are moved at the pins.

18. The device of claim 12, wherein the hook is configured with a broad tip which slides over the flat section of the top surface of the first arm and the second arm when the hook is inserted into the second section of the main body; wherein the first arm and the second arm are engaged with the hook by resting above the broad tip to prevent the hook from sliding out of the second section; and wherein the hook is disengaged when the first arm and the second arm are moved away from the broad tip of the hook freeing the hook to slide out of the second section.

19. The device of claim 14, wherein a lower end of the lock piece, facing toward a center wall, includes a projecting element having a channel facing toward the closed bottom, and wherein the center wall has a central groove with a knob included on the closed bottom within the central groove, wherein the projecting element of the lock piece slides into the central groove with the knob accommodating the channel of the projecting element when the lock piece slides into the first section.

20. The device of claim 19, wherein the first arm and the second arm each have a wide tip wherein each of the wide tips extends toward the central groove in the center wall to engage with the projecting element of the lock piece, wherein the wide tips extending above the projecting element prevent the lock piece from moving out of the first section; and wherein the lock piece is disengaged when the wide tips of the first arm and the second arm are moved away from the projecting element freeing the lock piece to slide out of the first section.
